

1 Motivation

As we see from the paper, TNBC is a highly heterogeneous subtype, hence spatial context is critical for elucidating tumor-immune-stromal interactions. Using Non-Negative Matrix Factorization (NMF) with Spatial Transcriptomics (ST), helps deconvolve mixed transcriptional profiles from spatial spots into latent programs (e.g., tumor, immune, stromal), providing interpretable gene loadings and quantitative spatial weights.

2 Methods

2.1 Data

- **Expression matrix:** m spots $\times$ n genes, exported from *Robust*
- **Coordinates:** $\{(x_j, y_j)\}_{j=1}^m$, pixel coordinates registered to the H&E image.

2.2 Preprocessing & Filtering

- **Highly variable genes:** top 2,000 by spot-wise variance, capturing true biological heterogeneity.
- **Log transformation:** $\log(1 + \text{counts})$ to stabilize variance.

2.3 Model Evaluation

- **Elbow analysis for choosing optimal value of k :** Run NMF for $k \in [2, 10]$, plot reconstruction error versus k , and select the “elbow” point where additional factors yield diminishing returns.

Just to verify, if we should use $k=3$, or 4, we calculated % variance explained based on Frobenius Norm as follows:

- **Reconstruction error:**

$$\|X - WH\|_F$$

where $\|\cdot\|_F$ denotes the Frobenius norm.

- **Variance explained:**

$$\%Var = 100 \times \left(1 - \frac{\|X - WH\|_F^2}{\|X\|_F^2}\right).$$

We get:

% variance explained for $k=3$: 92.8%

% variance explained for $k=4$: 93%

We went ahead with $k=3$ as it explains almost as much variance as with $k=4$. But with $k=3$, found 1 of the factors corresponding to housekeeping genes. So, we switched to $k=4$, and skipped the factor corresponding to housekeeping genes.

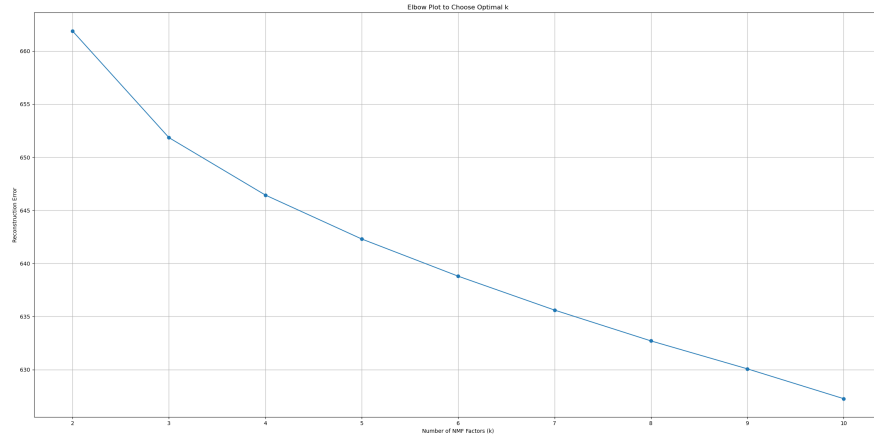


Fig. 1 Reconstruction error vs. number of NMF factors, k

2.4 Spatial Trend Analysis

1. **Spearman correlation** of each factor's weights $W[:, i]$ with x and y .
2. **2D scatter maps**: spot-level overlays colored by factor weight.
3. **Quiver plots**: gradient vector fields.

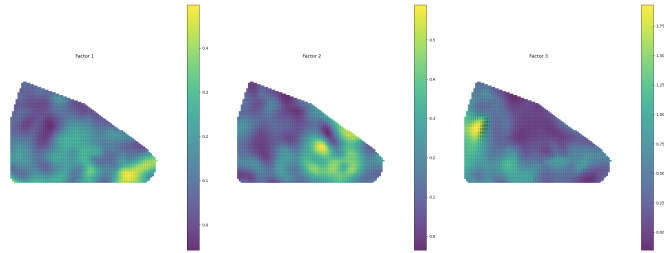


Fig. 2 Factors as interpolated heat map

2.5 Histology Overlay

Factor weights and gradient arrows are overlaid on registered H&E to integrate transcriptomic programs with morphology.

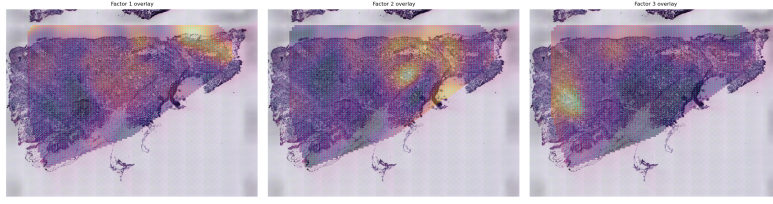


Fig. 3 Overlay of Factors as heatmap on H&E.

3 Results

3.1 Factor Interpretations

Factor	Top Genes	Spatial Trend	Interpretation
1	CD74, IGKC, B2M, HLA-B	increasing along Y ($\rho = 0.66$)	Immune
2	SCGB2A2, MGP, P4HB, EPHB3	increasing along X ($\rho = 0.62$)	Stromal/Secretory
3	CD24, TACSTD2, GLYATL2	Mixed (weak)	Tumor

Table 1 Summary of factor gene signatures, spatial trends, and interpretations.

4 Discussion

NMF at $k = 4$ and skipping the factor corresponding to (MT, RPS/RPL, MALAT1/ACTB/GAPDH), cleanly separates immune, stromal, and tumor programs. Gradient analyses reveal clear directional patterns—immune infiltration along the vertical axis and stromal polarization horizontally.